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Long overdue

Coal exchanges can balance out energy scarcities across India

Unveiled at a time of record domestic coal production, the **Coal Exchange Rules, 2026**, are a case of better-late-than-never. They will create a broad market-based mechanism through regulated trading platforms for the lynchpin of India's energy system – coal. They are aimed at enhancing price discovery, transparency, access for small consumers, as well as, one would hope, reduce bilateral agreements that are often opaque and come with a whiff of graft, too often. Today, most coal transactions between producers and buyers take place through long-term contracts, primarily for the power sector, followed by auctions, imports and captive mining. While India's commodity exchanges are well established, they function largely as financial markets rather than physical delivery platforms. Coal exchanges, however, appear closer in design to power exchanges, which, despite modest volumes, play a role in price discovery, market signalling and the development of secondary markets. As if to prove this point, coal exchanges are expected to serve the non-regulated sector, which relies on Coal India auctions where coal is often sold at a premium to the highest bidder. Power exchanges are not merely niche trading platforms; they serve as points of reference for the broader power market. They have enhanced price discovery and served as a balancing market without replacing long-term power purchase agreements. Initially the power exchanges were only balancing shortages, but eventually the spot prices became a barometer of the broader power market indicating scarcity, surplus and system stresses for all electricity stakeholders. Perhaps the first role of coal exchanges could be to open up inventories, allowing surpluses to balance out shortages across India.

The templates for the two exchanges are not very different though the specific rules framed by the **Coal Controller Organisation of India** will determine the success of coal exchanges. Just as with the successes, the failures of power exchanges can also serve as lessons learned for coal. Coal is not as fungible as electricity, which once generated is the same everywhere requiring only minimum standards. Coal quality varies widely. Therefore, robust standards and quality assurance are as important as contract design, liquidity creation and enforcement. The latter set of requirements will ensure that major producers and consumers are drawn to the coal exchanges. The emphasis should be on facilitating participation of retail consumers unlike power exchanges, which are dominated by discoms. Coal India's stance will be crucial. Besides safeguards against volatility, dispute resolution mechanisms and improved transportation logistics will be important too, since the coal exchanges will be physical delivery platforms.

Subject: Economy

Coal Exchange Rules 2026

The rules provide the legal and operational framework for establishing and regulating **coal exchanges** in India—similar to commodity or stock exchanges, but for coal trading. These exchanges will allow buyers and sellers to trade coal transparently through electronic platforms.

Key features include:

- Establishment of regulated electronic coal exchanges
- Trading through standardized contracts
- Market-based price discovery
- Registration and regulation by the **Coal Controller Organisation (CCO)**
- Participation by commercial miners, captive miners, and even public sector companies
- Provisions for settlement, governance, surveillance, and dispute handling

The reform is significant because India is:

- the world's second-largest coal consumer,
- heavily dependent on coal for electricity generation,
- and trying to improve domestic coal efficiency while reducing imports.

For UPSC preparation, this topic is important under:

- **Indian Economy**
- **Energy Sector Reforms**
- **Mineral Policy**
- **Market Reforms and Governance**

Feature	Coal Exchanges	Commodity Exchanges	Power Exchanges
Main Product Traded	Coal	Multiple commodities (gold, crude oil, wheat, cotton etc.)	Electricity
Nature of Product	Physical mineral fuel	Agricultural, metal, or energy commodities	Intangible energy flow
Delivery Requirement	Usually physical delivery of coal	Physical delivery or derivatives/futures	Real-time electricity supply
Trading Horizon	Spot and possibly forward contracts	Spot, futures, options	Day-ahead, real-time, term-ahead
Primary Objective	Transparent coal distribution and pricing	Hedging, speculation, price discovery	Efficient power balancing and grid optimization
Key Users	Coal producers, power plants, steel/cement industries	Traders, producers, investors, hedgers	Power generators, DISCOMs, industrial consumers
Regulator in India	Coal Controller Organisation (CCO)	Securities and Exchange Board of India (SEBI)	Central Electricity Regulatory Commission (CERC)

Indian industry is
at a crossroads. It
needs to look at
longer horizons



THE FIRST part of this article traced four structural ways of why chronic underinvestment in research and development. Apart from them, two further explanations result — one sociological, one generational — before the case for a different calculus.

The fifth explanation is chronological. The last 15 years have been among the most volatile in living memory. The financial crisis of 2008, the Eurozone debt crises through 2015, India's own oil-price shock including as the infrastructure boom slowed, the Covid lockdowns, the Russia-Ukraine war's supply and energy shocks, and now the Trump administration's tariff turbulence — each arrived before the last had fully receded. Firms have refused to commit to decade-long R&D programmes in such an environment to ask them to plant orchards in a forest fire. The rational response to compounding uncertainty is to preserve liquidity and defer irreversible commitments. This does not excuse the deficit in Indian R&D; it contextualises it.

A more intimate explanation lies alongside the macroeconomic crises. Indian business, like business elsewhere, is predominantly family owned. Family firms tend to follow a recognisable trajectory. The founder builds with hunger; the second generation consolidates, and by the third, the urgency has often dissipated. Mike Wollman and Wang in the *Journal of Economic Perspectives* in 2006, documented the tendency of multigenerational control toward economic retrenchment — preserving inherited wealth rather than risking it — while Peter Corcoran, working CIO succession in 2006, found performance and investment efficiency declining when a family heir took over rather than a professional. In India, this has acquired a post-pandemic inflection. The exceptional returns of financial markets have given third-generation families an effortless alternative to the capital-intensive work of manufacturing and R&D. The aggregate consequence is a quiet withdrawal of entrepreneurial energy from the domain India where most needs it.

No course, however illuminating, cannot be mistaken for an occupation. Indian industry stands at a juncture where the choice between short- and long-horizon thinking is no longer merely a matter of competitive positioning — it has become a question of strategic survival. India's ambitions for influence in a multipolar world depend on the technological and productive capabilities of its private sector. Strategic leverage in the 21st century is not purchased with diplomatic skill alone. It is underwritten by industrial capacity, by the ability to offer the world goods and services that others cannot readily replicate, and a position in global value chains that confers genuine bargaining power.

A private sector that captures domestic demand but remains technologically dependent — a creature of protectionist policy rather than a generator of it — is one whose future profitability is under the most accumulating existential threat. As competition from East Asia continues to climb the value chain, the space for Indian businesses that decline to invest in frontier capabilities will narrow. The business community has historically looked to the state to create conditions for prosperity. The state's obligations in this regard are not small. But the capacity for strategic influence is not the state's to create unilaterally. It depends on decisions taken in competitive landscapes.

About R&D budgets, talent, the willingness to absorb short-term costs in pursuit of long-run competitive advantage, national and shareholder interests converge in the long run. The private sector that fails to recognise this convergence will eventually discover it the hard way.

History, however, is not without encouragement, and the past term of transformation is not unprecedented. Germany's Mittelstand — the dense ecosystem of mid-sized manufacturers with deep engineering capabilities — was forged through deliberate investment in technical education, patient capital from Hausbank relationships that insulated firms from quarterly market pressures, and a cultural commitment to *Qualität*. Japan's post-war industrial miracle rested on long-horizon corporate strategies, state-guided but privately executed investment in technological capability, and a willingness to absorb short-run losses in pursuit of competitive positions in the industries that would matter in the future. South Korea's chaebols, for all their well-documented governance imperfections, developed world-class semiconductor and shipbuilding capabilities through a willingness to invest at scale and over a duration that would have been impossible under the tyranny of the quarterly earnings cycle.

Each of these transformations was preceded by a recognition, often born of crisis, that the existing trajectory was insufficient and that the path to national prosperity ran through industrial capability not amenable to India's arguably such a moment. The fracturing of the old globalisation, the imperative and the difficulty of supply-chain diversification away from China, the unprecedented window of opportunity a demographic dividend that is available but will not remain so indefinitely — these are not guarantees of transformation, but conditions that make it possible. The short horizon has served Indian business reasonably well in the long run of low-hanging domestic demand. The next one will belong to those who learn, perhaps not too late, to look further.

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Subject: Economy

What are the two additional explanations (chronological and generational) for India's R&D underinvestment beyond the four systemic causes?

1. Chronological Explanation

This explanation argues that the **last 15 years have been exceptionally volatile**, discouraging long-term investments like R&D.

Major shocks included:

- the 2008 global financial crisis,
- Eurozone debt crisis,
- India's infrastructure and banking stress,
- COVID-19 pandemic,
- Russia-Ukraine war,
- global supply-chain disruptions,
- and renewed tariff uncertainties.

Because R&D investments usually take many years to generate returns, businesses became cautious in such an unstable environment. Firms preferred:

- preserving liquidity,
- avoiding irreversible commitments,
- and focusing on short-term survival.

The article compares this situation to:

“planting orchards in a forest fire.”

Thus, chronic uncertainty increased risk aversion and delayed innovation spending.

2. Generational Explanation

This explanation relates to the structure of Indian business ownership, especially the dominance of **family-run firms**.

The argument follows a common pattern:

- **First generation** builds the business with ambition and risk-taking,
- **Second generation** consolidates and stabilizes it,
- **Third generation** often becomes more conservative and wealth-preserving.

As businesses mature across generations:

- appetite for risky innovation may decline,
- focus shifts toward protecting existing wealth,
- and long-horizon R&D investments become less attractive.

Since R&D requires:

- patience,
- uncertainty tolerance,
- and delayed returns,

family firms may prefer safer investments over experimental

Germany, Japan, and South Korea succeeded because they:

- prioritized long-term national capability,
- invested patiently in technology,
- aligned industry with state strategy,
- and accepted delayed returns for future competitiveness.

long-term thinking is no longer merely about becoming richer — it is about remaining economically sovereign, strategically relevant, and globally competitive in the 21st century.

The 8th CPC — a chance to reform pay commissions

As India moves toward the 8th Central Pay Commission (CPC), public discussion has largely centred on familiar themes – fitment factors, salary revisions and arrears. Yet, the larger question may not be how much compensation should increase, but whether the framework for determining public compensation remains coherent, equitable, and fiscally sustainable. This is not merely an administrative matter. The manner in which the state structures salaries, allowances and pensions reflects broader institutional priorities and influences public confidence in governance.

The framework deficit

Over the decades, Pay Commissions have become far more than wage-revision exercises. Their recommendations shape inter-service parity, long-term fiscal commitments and the institutional balance within the state. Yet, the process remains narrow, with a small, time-bound body tasked with evaluating a diverse ecosystem of civil, military and technical services, largely on the basis of representations from the services themselves. A key challenge is the absence of a common evaluative framework. Services operate under different structures, career trajectories and working conditions, but there is no universally accepted method for comparing risk, responsibility, technical complexity or career progression. As a result, the system often seeks parity without clearly defining its basis.

This challenge is most visible in questions of parity. Officers across services with very different career paths and responsibilities may, at certain stages, receive comparable compensation. The issue is not one of competition between services but institutional coherence. If parity is to be maintained, the principles governing it must be transparent, consistent and objectively justified.

The comparison between civilian services and the armed forces illustrates this complexity. Military careers follow a sharply pyramidal structure, with limited promotional opportunities



Prem Kumar Nair

Retired Indian Army officer (colonel) with an interest in governing architecture and systems

and earlier retirement. Civilian services generally offer broader avenues for advancement and longer careers. Aligning compensation across such different systems requires careful consideration of these structural differences.

Another significant trend is the reduction in experience required for senior administrative positions. While faster career progression may reflect changing governance needs, complex policy challenges still demand institutional memory and seasoned judgment. Efficiency is important, but it cannot fully substitute for experience. Questions also arise over allowances. While they are intended to compensate for hardship, remoteness or operational risk, there is no uniform and transparent framework for assessing such conditions across services. This can create disparities that are difficult to justify and may foster perceptions of inconsistency.

Non-Functional Upgradation (NFU) raises a related concern. By allowing financial advancement without a corresponding increase in responsibility, it weakens the link between role, accountability and compensation. Though introduced to address slower promotion avenues, the NFU continues to generate debate over equity and institutional rationale.

The pension challenge

Pensions add another layer of complexity. India currently operates multiple pension systems, including legacy defined-benefit schemes, contributory plans for newer entrants, and separate arrangements for elected representatives. According to the Reserve Bank of India's State Finances Report (2023), salaries, pensions and interest payments consume a large share of State expenditure, limiting fiscal space for development. As a result, concerns about sustainability and inter-generational equity are becoming increasingly important.

Taken together, these issues highlight the need for a more coherent approach to public compensation. At present, pay frameworks for the executive, legislature and judiciary evolve

through different processes. While constitutionally distinct, this fragmentation can create inconsistencies and reduce transparency in compensation decisions.

Public trust depends not only on fairness, but also on transparency and explainability.

A new compensation architecture

Many countries have gradually moved towards continuous and institutionalised mechanisms for reviewing public sector compensation. Independent authorities, clearly defined benchmarks and periodic review mechanisms have replaced infrequent, large-scale revisions. India's decadal Pay Commission model may therefore warrant reconsideration.

A more durable framework – whether a **National Compensation Authority** or a specialised public service body – could bring greater consistency to public sector pay. Rather than be a centralising authority, it could establish common principles for assessing responsibility, experience and hardship while preserving flexibility for different services and States.

Importantly, any such reform must respect India's federal structure. States should retain autonomy over implementation while operating within a broader framework of transparency, comparability and fiscal discipline. Greater coherence across branches of government, without affecting constitutional independence, would strengthen both credibility and public confidence.

Ultimately, public compensation is not simply about salaries and pensions. It is part of the larger relationship between the state and the citizen. In a democratic system, compensation structures must not only be financially sustainable, but also publicly explainable.

The 8th Pay Commission presents an opportunity to move beyond periodic revision and engage with these deeper structural questions. Whether that opportunity is utilised meaningfully may shape public trust in institutional governance for years to come.

Subject: Polity & Governance

labour markets change rapidly, technology alters job structures, private-sector compensation evolves continuously, and pension liabilities are expanding.

The 8th Central Pay Commission should reform compensation frameworks, not merely salaries

Key Structural Challenges the 8th Central Pay Commission (8th CPC) Must Address

Inter-Service Parity

Defence vs Civil Services Tension

Pension Burden

Old Pension Scheme (OPS) for older employees
National Pension System (NPS) for newer recruits

Other Structural Issues the 8th CPC May Need to Address

Performance vs Seniority

Talent Competition with Private Sector

Fiscal Sustainability

Case for a Permanent National Compensation Authority (NCA)

Continuous Review Instead of 10-Year Shocks

Professionalized Compensation Management

Dynamic Labour-Market Benchmarking

Faster Response to Emerging Sectors
International Practice

Equality of treatment for Persons with Disabilities

Addressing the digital divide through the Digital India Mission, India has marched on towards becoming a global "best practice" model of a digital welfare state. Yet, Persons with Disabilities (PwDs) remain largely excluded from this promise of universal outreach. Disability pensions, contrary to the very principles underlying disability rights, are determined not by the nature or extent of disability but by domicile, the discretionary decisions of State governments, and cumbersome bureaucratic processes. As a result, a vulnerable section of citizens continues to remain outside the ambit of a welfare architecture that otherwise prides itself on inclusivity and last-mile delivery.



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An inadequate safety net

The 2011 Census recorded 2.68 crore PwDs. Today, accounting for population growth and changing disease profiles, their number is conservatively estimated at 4.5 crore-6 crore. Although the Supreme Court of India has recognised the right to live with dignity as a fundamental right and the Rights of Persons with Disabilities Act, 2016 provides legal protection, disability benefits – especially pensions – remain fragmented, discretionary, and inadequate. As a result, the Indira Gandhi National Disability Pension Scheme covers only a small fraction of PwDs, while pension amounts in most States range from just ₹300 to ₹500 a month, with a few offering ₹1,000-₹3,000.

India spends barely 0.02% of GDP on disability welfare including pensions. South Africa spends 0.12%-0.15% of GDP (six times more); Brazil 0.45%-0.50% (20 times more); OECD countries 2.2% (110 times more) and Australia 0.35%-0.40% (20 times more).

Beyond politics, sound economics demands the inclusion of PwDs. The World Bank and UNDP estimate that low- and middle-income countries lose 3%-7% of GDP when PwDs are excluded from education, employment, and social security. Disability income improves household stability, rural consumption, and labour participation. Studies show fiscal multipliers of 1.4-1.6, while the 2025 Pro Bono Economics report found that the socio-economic returns from disability pensions exceed their costs by nearly 48%. Far from being a welfare expense, disability pensions are an investment and an effective economic stimulus.

India must establish a Minimum Universal Disability Pension Floor Rate (MUDPFR) to give effect to the state's constitutional obligation, under Article 41, to provide public assistance to persons with disabilities, and to operationalise Section 24 of the Rights of Persons with Disabilities Act, 2016, which guarantees adequate

India must
adopt a
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pension
floor rate

social security, including pension benefits. Such a measure would translate the promise of inclusive growth and "Sabka Saath, Sabka Vikas" into reality by shifting disability pensions from a matter of charity and discretion to a matter of citizenship rights. While ensuring that no disabled person receives less than a minimum pension regardless of where they live, a MUDPFR would still allow States to provide additional top-ups.

This is not an unrealistic proposal. South Africa provides a national disability grant with uniform eligibility norms; Brazil's BPC guarantees a national minimum income; and Australia and New Zealand operate nationwide disability pension systems. Several developing countries, including Kenya, Rwanda, Thailand and Indonesia, also provide disability income support at the national level. International experience shows that centrally set standards promote uniformity, universality and portability.

A MUDPFR of ₹8,000 per month for 40 lakh beneficiaries would cost about ₹38,400 crore annually (0.08% of GDP), while ₹10,000 for 65 lakh beneficiaries would cost ₹78,000 crore. Even a pension of ₹15,000 per month would keep expenditure below 0.2% of GDP. Such spending is fiscally manageable when compared with allocations for food subsidies (₹2.05 lakh crore), rural development (₹1.80 lakh crore), tax concessions and revenue foregone (₹1.72 lakh crore), and infrastructure (₹11.1 lakh crore). Yet, disability pensions and inclusion continue to receive only a tiny fraction of public expenditure.

From fragmentation to integration

The current disability pension system, shared between the Ministry of Rural Development and the Department of Empowerment of Persons with Disabilities, leads to duplication, delays and diffused accountability. Countries that have faced similar challenges have addressed them through a single national authority – South Africa's SASSA, Australia's NDIA, Brazil's INSS and Ireland's Department of Social Protection. India too needs a National Disability Pension Authority to oversee eligibility norms, a national registry, portability, digital integration, grievance redress and State-wise performance monitoring – one standard, one system, one nation.

Implementing a robust disability pension system would strengthen India's bid for a UN Security Council seat by translating its commitments into action under Article 28 of the UN Convention on Rights of Persons with Disabilities, ILO Recommendation No. 202, SDG 1.3, and the G-20 New Delhi Leaders' Declaration, all of which endorse social protection as a cornerstone of inclusive development.

Combining pensions with employment

support can move persons with disabilities from mere survival to productive participation. A MUDPFR could also yield economic benefits. Countries such as Singapore, South Korea, South Africa and Brazil integrate disability pensions with employment and social security systems. India should similarly strengthen its fragmented Disability Employment Incentive Scheme. Models abroad include employer tax incentives in Nigeria, the U.K.'s Access to Work programme, and Australia's wage subsidies. Existing schemes such as PM-DAKSH, NAPS and State-level employer incentives provide a foundation for expansion.

Constitutional imperative

A MUDPFR would give effect to the Constitution's guarantees of equality, equal protection, dignity and the right to life. India has already standardised major welfare programmes – from food security and health care to PM-KISAN and pensions – and delivers benefits at scale through DBT and UPI. The capacity and technology exist. What is needed is the political will to prioritise dignity. Disability pensions remain one of the few entitlements determined by where a person lives. If dignity is a constitutional right, geography cannot decide the minimum support for survival.

More than an economic or administrative reform, a MUDPFR is a moral one. It affirms that persons with disabilities are rights-bearing citizens, not recipients of charity. By transforming the state from a benevolent provider into a constitutional guarantor, it strengthens dignity, inclusion and citizenship.

Should India continue with a system in which disability pensions vary according to State budgets, political priorities and bureaucratic complexities, leaving persons with disabilities vulnerable to hardship? Or should the Republic guarantee a minimum level of support for all? A Viksit Bharat cannot leave its most vulnerable citizens at the mercy of a postcode lottery. Federalism cannot be a justification for inequality.

A MUDPFR is the next logical step in India's welfare architecture, building on rights-based entitlements enabled by digital inclusion and DBT. It would humanise the state, uphold the dignity of persons with disabilities and strengthen the Republic.

Over the past 75 years, India has steadily expanded its welfare state in pursuit of social justice. With the necessary platforms and delivery systems now in place, the time has come to secure the foundation. The real question is whether India recognises dignity for persons with disabilities as a constitutional right and a collective obligation. That promise is long overdue, and a MUDPFR offers a way to fulfil it.

Social Justice

The **Minimum Universal Disability Pension Floor Rate (MUDPFR)** is a proposed policy framework in India that seeks to guarantee a **national minimum disability pension** for every eligible Person with Disability (PwD), irrespective of the State in which they live.

1. Huge interstate disparities
2. Current pensions are inadequate
3. Rights-based approach instead of charity
Article 14 — Equality before law,
Article 21 — Right to life with dignity,
Article 41 — Public assistance in cases of disability,
Section 24 of the Rights of Persons with Disabilities Act, 2016

4. Growing disability population
5. Economic inclusion

MUDPFR represents a shift:

- from **welfare to rights**,
- from **fragmented state schemes to universal minimum protection**,
- from **survival assistance to inclusive development**.

India's current disability welfare architecture — especially the **Indira Gandhi National Disability Pension Scheme (IGNDPS)** under the National Social Assistance Programme (NSAP) — suffers from several structural, financial, and administrative gaps.

- 1. Extremely low pension amount**
- 2. Exclusionary eligibility criteria**
- 3. Dependence on outdated BPL data**
- 4. Interstate inequality**
- 5. Bureaucratic and certification hurdles**
- 6. Weak implementation and monitoring**
- 7. Fragmented institutional architecture**

a **Minimum Universal Disability Pension Floor Rate (MUDPFR)**,
universal disability income support,
portability,
simplified certification,
and a rights-based national disability
protection framework.

- 8. Welfare-centric rather than rights-based approach**
- 9. Poor coverage and statistical invisibility**
- 10. Lack of integration with employment and social inclusion**

Why Zojila tunnel breakthrough is a boost for military mobility

Bashaarat Masood
Srinagar, June 12

THE ZOJILA tunnel, the world's longest high-altitude bi-directional road tunnel, is a step closer to completion after a "breakthrough" was achieved in its construction on Tuesday — marking the end of the excavation phase. The tunnel, which will connect Kashmir Valley and Ladakh throughout the year, is one of India's most important strategic infrastructure projects.

The project is significant for residents, who can soon expect all-weather connectivity, and for the armed forces, who can now reduce their dependence on air transport in the strategic Himalayan region.

The tunnel

The Zojila tunnel, situated at an altitude of 11,758 feet, is a 13-km road tunnel connecting Baltal (near Sonamarg) in Kashmir valley and Minimarg (near Drass) in Ladakh's Kargil district.

The tunnel bypasses the treacherous Zojila pass — a narrow, winding road at an

elevation of 16,430 feet — that remains closed in the winters and is prone to landslides throughout the year. This will reduce the distance from Sonamarg to Drass by 27 km and cut travel time by more than two hours. The tunnel, which is now targeted to be completed in February 2028, forms the centrepiece of the 30.894-km Zojila project. It includes roads, bridges and tunnels aimed at ensuring year-round access to Ladakh.

All-weather connectivity

The Union Territory (UT) of Ladakh is connected to the rest of the country by two main highways: the Srinagar-Sonamarg-Leh highway and the Leh-Manali highway. Both roads, however, remain closed for almost six months every year owing to snow, avalanches and extreme weather.

People in Ladakh have long been demanding a road tunnel that connects the UT to Kashmir Valley. Besides fulfilling that demand, the tunnel will also provide swift round-the-year mobility to defence forces in one of India's most sensitive border regions.

An all-weather link is especially impor-

Project delays

● The Rs 6,800 crore project, whose foundation stone was laid in May 2018, was expected to be completed by September 2026.

● But the Covid-19 pandemic, the 2024 terror attack on the workers at the Sonamarg tunnel, and extreme weather conditions delayed it.

tant for the forces, as it will boost manoeuvrability in case of hostilities with China on the Line of Actual Control in Ladakh or with Pakistan on the Line of Control in Kargil.

The all-weather road is a strategic lifeline that will facilitate the seamless movement of personnel, military hardware, fuel and daily supplies — ending the long-standing reliance on expensive and vulnerable airlifts. It is expected to significantly enhance military mobility by enabling quicker deployment of troops and equipment and reducing response times in the event of hostilities. It will also offer greater protection from enemy surveillance and artillery fire, vulnerabilities that were evident during the 1999 Kargil War when the key road link came under artillery fire.

The other key tunnel that keeps the Srinagar-Leh highway operational throughout the year — the Z-Morh tunnel or Sonamarg tunnel — was completed and thrown open for traffic in January last year.

What next

The Zojila tunnel project has several key

components. Besides the main Zojila tunnel are two smaller ones — one 457 metres long and the other 1,953 metres long — at Nilggar.

The larger Zojila project also comprises four bridges, an approach road from Sonamarg tunnel to Zojila tunnel, three vertical shafts — structures that provide ventilation and emergency access to the tunnel as it has no separate escape tunnel — and seven "cut-and-cover" (tunnel-like) structures. Most of these projects have already been completed.

"What is remaining is the balance works in the vertical shafts, we have already made the hole and have to widen 6.6 metres. It is the most challenging job we are up to and we expect it to complete in one-and-a-half years," said Colonel A K Shivkumar, Zojila tunnel's Project Manager.

"We also have to do the balance work for the main tunnel. We have completed the inner concrete lining for 3 km and the rest of the stretch is expected to be completed in a year. In addition to this, we have to build a drainage system and set up a ventilation system, electrical system and a fire-fighting system," he added.

The tunnel is part of the larger **Srinagar-Leh highway modernization programme** on National Highway-1 (NH-1). At present, the Zoji La pass remains snowbound for several months every year, often cutting off Ladakh from the Kashmir Valley during winter.

- ensure **year-round road connectivity**,
- reduce travel time,
- improve civilian logistics,
- strengthen military deployment capability near both the:
 - Line of Actual Control (LAC), and
 - Line of Control (LoC).



How the Tunnel Enhances India's Military Mobility

1. All-weather strategic access to Ladakh
2. Faster response near the LAC
3. Strengthening the Kargil–LoC sector

Lessons from the 1999 Kargil War

1. **Vulnerability of a single seasonal highway**
NH-1A (now NH-1) was India's lifeline to Kargil and Leh
2. Logistics decide mountain warfare
3. Infrastructure gaps create strategic asymmetry
4. Need for rapid mobilization against a two-front challenge

Strategic Significance

The Zojila Tunnel converts Ladakh from a **seasonally connected frontier** into a **year-round militarily sustainable theatre**, fundamentally improving India's strategic posture in the Himalayas





Key specifications

- **Length:** ~14.15 km
- **Altitude:** around 11,500 feet (3,500 m) above sea level
- **Type:** bi-directional single-tube road tunnel with escape provisions
- **Highway:** NH-1
- **Design speed:** about 80 km/h
- **Travel time reduction:** from 3–3.5 hours over the pass to around 15–20 minutes
- **All-weather access:** designed for uninterrupted winter movement
- **Tunnel width:** approximately 9.5 m carriageway
- **Safety systems include:**
 - ventilation systems,
 - fire safety mechanisms,
 - emergency escape routes,
 - avalanche protection structures,
 - intelligent traffic

Thank you

“The price of success is daily effort”



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